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Acknowledgments:

SH thanks Chantal Ripp for assistance with industry profit margins and Margaret Rose for help collecting revenue and profit data of major companies.

Funding:

This work contributes to ICTA-UAB "María de Maeztu" Programme for Units of Excellence of the Spanish Ministry of Science and Innovation, CEX2024-001506-M/funded by MICIU/AEI/10.13039/501100011033, (DB).

Volkswagen Foundation grant 9C784 "Repercussions of Open Access on Research Assessment (ROARA)" (SH)

Author contributions:

Conceptualization: FB, DB, PC, GD, AF, PGB, MAH, SH, VL, CN, SP, JW

Investigation: DB, SH, VL

Visualization: PC

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Writing: FB, DB, PC, GD, AF, PGB, MAH, SH, VL, CN, SP, JW

Competing interests:

Authors declare that they have no competing interests.

Corrections:

Version 1 of this paper (released 6th Nov 2025) stated that publisher profits were over \$12bn between 2019-2024, when they are over \$14.7bn. We also under-estimated the NSERC budget.

This second version was released on 17th November 2025.

Drain of Scientific Publishing: Supplementary Materials

Materials and Methods: Sources for publishers' financial data

Wiley. 'Form 10k' annual reports divide revenues into 'Research', 'Publishing' and 'Solutions'. We included all revenues from 'Research'. We took their 'Adjusted EBITDA' data for this sector.

Springer-Nature. Revenues are broken down into four sectors: 'Research', 'Education', 'Health' and 'Professional'. Up until 2022, profits are only given for all four sectors combined. The 2024 report breaks profit down by sector for 2023 and 2024. We took the average contribution of Research in 2023 and 2024, to calculate profits for other years. Figures quoted are the Adjusted Operating Profits.

Taylor and Francis. Figures are included in the financial reports of its parent company 'Informa', which include revenues from the book publisher Routledge. Figures quoted are the Adjusted Operating Profit.

Elsevier. Annual reports of the parent company Relx include journals as part of their 'Scientific, Technical and Medical' sector as well as the data services that journals provide. We took the 'Adjusted Operating Profits' which are the only profit data provided at the sectoral level. EBITDA data are available, but only for the company as a whole. EBITDA profits tend to be higher than Adjusted Operating Profits. In 2024 for example, the adjusted operating profit for the company as a whole was US\$4.095 bn, whereas the EBITDA figure is US\$4.767 bn (2024 report page 198). It is possible therefore that the figures we report may under-estimate that drain on academic publishing that publishers' profits constitute.

Wiley reports financials in US dollars, Elsevier and Taylor and Francis in UK Sterling and Springer Nature in Euros. We used currency conversion data, taking the average exchange rate value for each calendar year from this site: www.exchangerates.org.uk. We then converted all data into dollars.

S1. Examples of alternative publishing practices

Latin America is a region that highlights for its extensive development of diamond publishing, where a virtuous value system is observed as a result of indexation systems that focus on academic quality and independent editorship. This noncommercial and community-led publishing circuit finds its main strength in its public nature and the crucial role of universities. 11,117 active journals are published today through 738 University Portals managed by centralized editorial teams, mostly using OJS. The resilience of diamond publishing is explained by this kind of institutional support. This should be further expanded because many diamond journals are at risk by the constant attempts to buy them by predatory companies and commercial publishers. As the editor of the Latin American Journal of Sedimentology said in an interview: "I receive at least one proposal per month to sell the journal, and the figures offered keep rising" (16).

In Tanzania, public universities and research institutions have budgetary allocations specific for supporting journals. Through the Consortium of Tanzania Universities and Research Libraries, 93 academic member institutions across the country have agreed to provide and support open access research information across the country. The consortium supports launching of open access publishing platforms, repositories and the adoption of open access

journal policies. The use of institutional repositories has particularly served as an important avenue for dissemination to the research communities, learners, and the general public. These coordinated efforts are new and statistics are not available for measuring the impacts. However, the University of Dar es Salaam (UDSM) - which is the oldest and largest public institution - had its 25 journals in Diamond Open Access until recently in 2024 when two of these entered into contract with Brill. This was locally perceived to be a measure of international recognition and so, perversely, welcome.

Thus, each region has different needs according to its specific context. But the commercial threats remain active everywhere and the prevalence of the dominant concept of “excellence” risks the future of diamond open access.

S2. Evidence of problems in scientific publishing

The multiple forms of strain and abuse in scientific publishing are increasingly being systematically documented (17-19). Here is a summary of the sources available.

Editorial board resignations. A list of board resignations is kept by Retraction Watch here: <https://retractionwatch.com/the-retraction-watch-mass-resignations-list/>

Systematic lists of problems can be found on pub-peer: <https://pubpeer.com/>

Also the group ‘United2Act Against Paper Mills’ is also mobilising work against harmful behaviour: <https://united2act.org/>

There are also commentaries and observations on publishers’ behaviour. For example, Springer-Nature launched a new series the ‘Discover’ series that so closely reassembled MDPI in its purpose and format that it even mimicked the names of MDPI journals. This prompted a humorous response from four of the present authors:

https://the-strain-on-scientific-publishing.github.io/website/posts/discover_nature/

And some of the mistakes are so bad as to be (almost) comical, where not the integrity of science to be at stake. Salient examples include the AI generated image of a massive rats penis ([here](#)), and the paper which used capital ‘T’s instead of error bars ([here](#)).

The problems are also visible in the action funders are taking. Gates foundation already demands all work be preprinted ([here](#)). The Swiss National Science Foundation have disallowed the use of their funding to pay the APCs of special issue articles ([here](#)). The Finnish Publication Forum decided to downgrade hundreds of MDPI and Frontiers journals because of quality concerns ([here](#)).

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Table S1. Comparative Financial Data**Academic publishers compared to 30 largest companies based on 2024 revenue**

Revenue and profit in million USD, from consolidated revenue. Companies sorted by profit margin.

	HEADQUARTERS	EMPLOYEES	REVENUE	PROFITS	MARGIN
Academic publishing · NAICS 511 · Mean industry net profit margin: 12%					
					Mean: 34%
RELX	United Kingdom	36,400	12,057	4,088	33%
↳ Elsevier	United Kingdom	9,700	3,899	1,497	38%
Informa	United Kingdom	11,400	4,542	1,271	27%
↳ Taylor & Francis	United Kingdom	11,000	892	327	36%
Springer Nature Group	Germany	9,092	1,998	554	27%
↳ Springer Nature Research Segment	Germany	6,125	1,529	488	31%
Wiley	United States	6,400	1,042	331	31%
MDPI	Switzerland	6,650	--	--	--
Frontiers	Switzerland	1,440	--	--	--
Information technology · NAICS 334 · Mean industry net profit margin: 27%					
					Mean: 29%
Microsoft	United States	228,000	245,122	88,136	35%
Alphabet	United States	183,323	350,018	100,118	28%
Apple	United States	164,000	391,035	93,736	23%
Oil and gas · NAICS 21111 · Mean industry net profit margin: 21%					
					Mean: 8%
Saudi Aramco	Saudi Arabia	75,118	480,446	106,246	22%
ExxonMobil	United States	60,900	349,585	35,063	10%
Chevron	United States	39,742	193,414	17,611	9%
TotalEnergies	France	102,887	241,550	18,264	7%
China National Petroleum Corporation	China	1,000,800	436,875	28,677	6%
Shell	United Kingdom	96,000	289,029	16,521	5%
China Petrochemical Corporation	China	355,952	428,286	20,805	4%
BP	United Kingdom	100,500	194,629	6,782	3%
Financials · NAICS 52 · Mean industry net profit margin: 19%					
					Mean: 34%
Industrial and Commercial Bank of China	China	415,159	109,507	51,115	46%
JPMorgan Chase	United States	317,233	177,556	58,471	32%
Berkshire Hathaway	United States	392,400	371,433	88,995	23%
Pharmaceuticals · NAICS 3254 · Mean industry net profit margin: 14%					
					Mean: 0%
CVS Health	United States	300,000	372,809	4,614	1%
McKesson	United States	51,000	359,051	3,481	0%
Cencora	United States	46,000	293,959	1,509	0%
Cardinal Health	United States	48,900	222,578	1,561	0%
Energy · NAICS 22111 · Mean industry net profit margin: 14%					
					Mean: 2%
State Grid Corporation of China	China	1,361,423	545,954	11,465	2%
Commodities · NAICS 213112 · Mean industry net profit margin: 13%					
					Mean: 0%
Vitol	Switzerland	1,560	331,000	--	--
Trafigura	Singapore	13,086	243,201	2,758	1%
Glencore	Switzerland	83,426	230,944	-2,694	-1%
Retail · NAICS 44 · Mean industry net profit margin: 7%					
					Mean: 4%
Amazon	United States	1,556,000	637,959	59,248	9%
Costco	United States	333,000	254,453	7,367	2%
Walmart	United States	2,100,000	642,637	15,511	2%
Construction · NAICS 23 · Mean industry net profit margin: 7%					
					Mean: 2%
China State Construction Engineering	China	382,894	315,588	7,558	2%
Automotive · NAICS 336 · Mean industry net profit margin: 6%					
					Mean: 6%
Toyota	Japan	380,793	297,628	32,636	10%
Volkswagen Group	Germany	679,500	351,342	20,626	5%
Stellantis	Netherlands	248,243	169,773	3,990	2%
Health Insurance · NAICS 52411 · Mean industry net profit margin: 3%					
					Mean: 3%
UnitedHealth Group	United States	400,000	400,278	14,405	3%

Industry net profit margins and industry classification obtained from Dow Jones Factiva Industry Snapshot. List of largest companies obtained from Wikipedia, revenues and profits (in million USD) and number of employees extracted from annual financial reports and converted to USD if necessary.

Source: https://en.wikipedia.org/wiki/List_of_largest_companies_by_revenue

Mean 2024 average exchange rates for USD used: GBP: 1.2781; EUR: 1.0822; RMB: 0.1393; JPY: 0.0066; NTD: 0.0312